

World Action Models are Zero-shot Policies

World Action Models: Imagine, Then Act

24th SEP

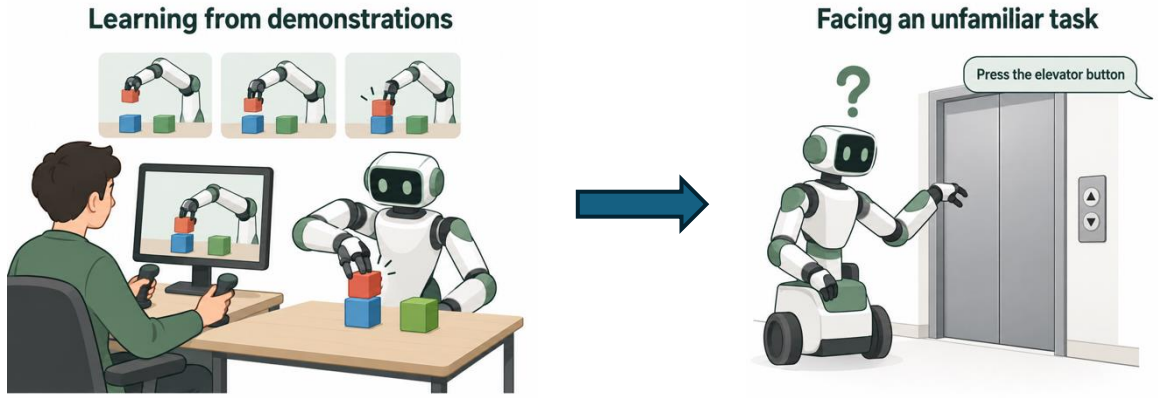
Shuai Wang

Background

Can a robot do a task it has never practiced?

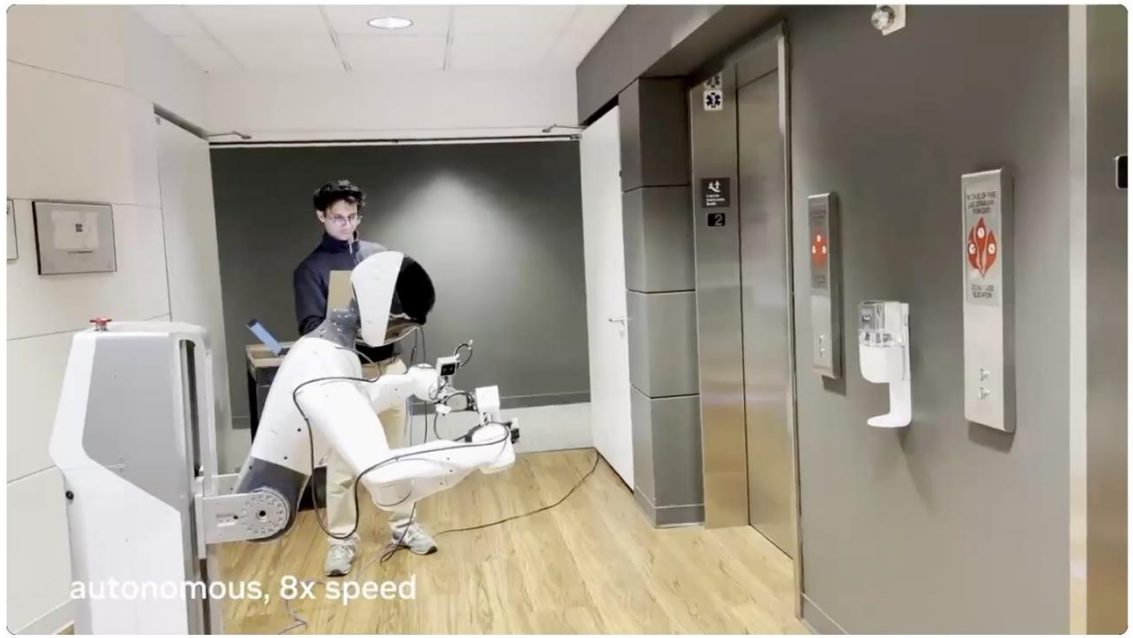
- **Background**

  Robots often learn tasks from demonstrations.
  Performing unfamiliar tasks remains a challenge.



- **Example**

  Move forward and press the elevator button with your right arm.



What Is a World Model?

➢ **Example**

A cup sits near the edge of a table.

What would happen if you pushed it further?

Current State



Possible Action



Predicted Outcome



You can imagine the outcome without actually trying it.

❖ A **world model** predicts what will happen next if we take an action.

- It predicts how an environment will change based on its current state.

Why Use a World Model?

❖ **World Model: Predict possible outcomes before acting.**

➢ Example: Move the cup to a safe position.

Possible action	Predicted outcome
Push the cup toward the edge	The cup may fall
Move the cup toward the center	The cup stays supported on the table



These predictions can help a planner choose an action.

➢ Where can this help?



Driving: Anticipate how a scene may change.

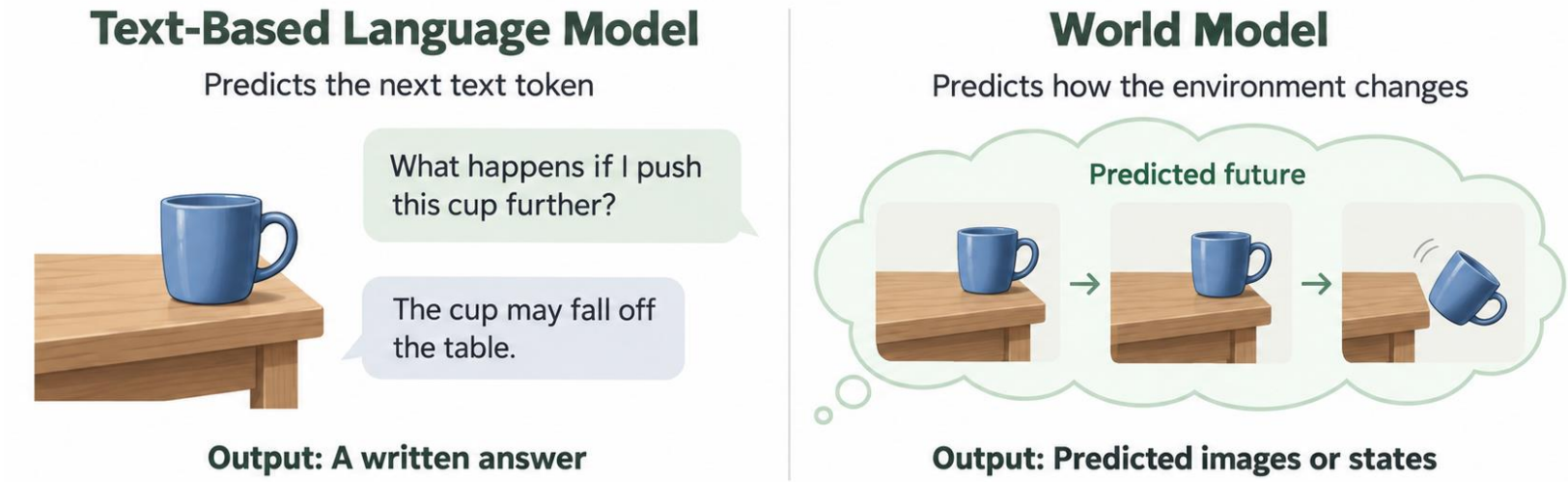
When driving, a system could predict:

- Could changing lanes now cause a collision?
- At this speed, can we avoid the pedestrian in time?

❖ Predict consequences before acting to help avoid risky actions.

World Models vs. Language Models

Prompt: What would happen if you pushed it further?



	Classic Language Model	A World Model for Physical Environments
Typical input	A question and text context	Observations and possible actions
Typical output	A written answer	Predicted images, states, or internal representations

Why Do We Need a World Action Model?

- A world model predicts future states.
  A robot also needs executable actions.

Example: "Move the cup away from the edge, toward the center of the table."

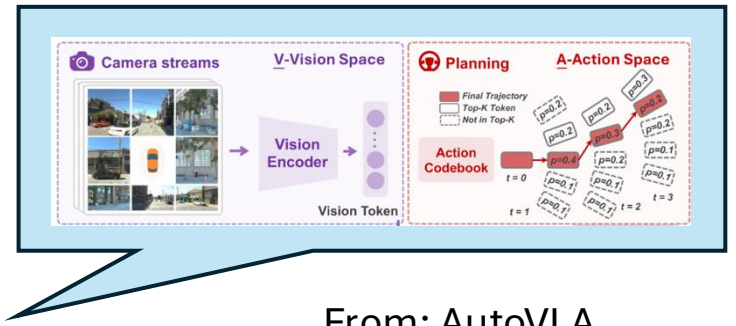
World Model
What happens next



+

Vision-Language-Action
What the robot should do





From: AutoVLA
[1] https://autovla.github.io/

World Action Model
Future states and actions

Proposed DreamZero



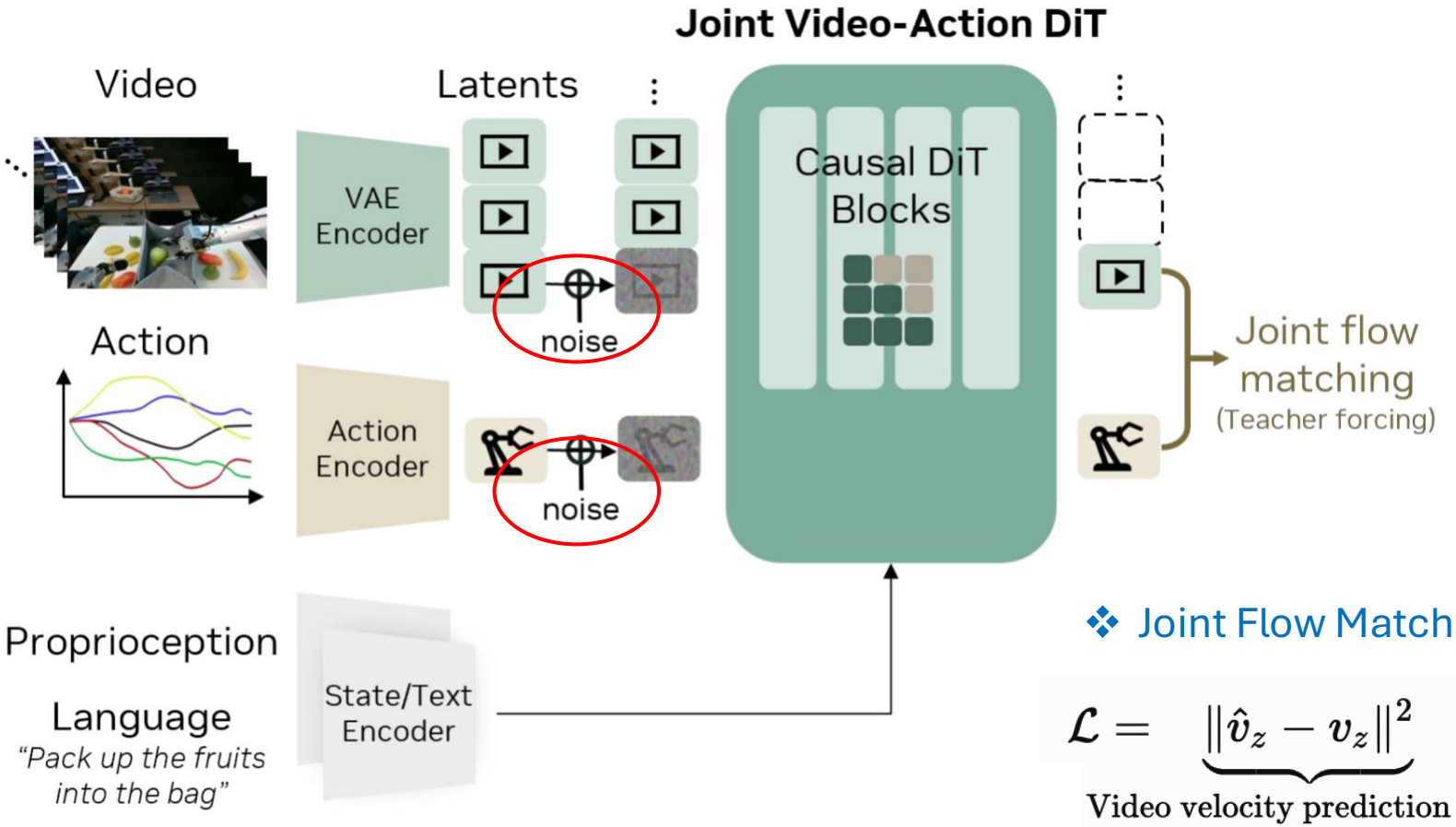
Future video and **robot action**

- Predict the movement. Generate the commands. Learn both together.

Method

Model Architecture of DreamZero

- Training: Joint video-action flow matching



❖ Joint Flow Matching Loss

$$\mathcal{L} = \underbrace{\|\hat{v}_z - v_z\|^2}_{\text{Video velocity prediction error}} + \underbrace{\|\hat{v}_a - v_a\|^2}_{\text{Action velocity prediction error}}$$

Advantage of such training

- **Learn from diverse experience**

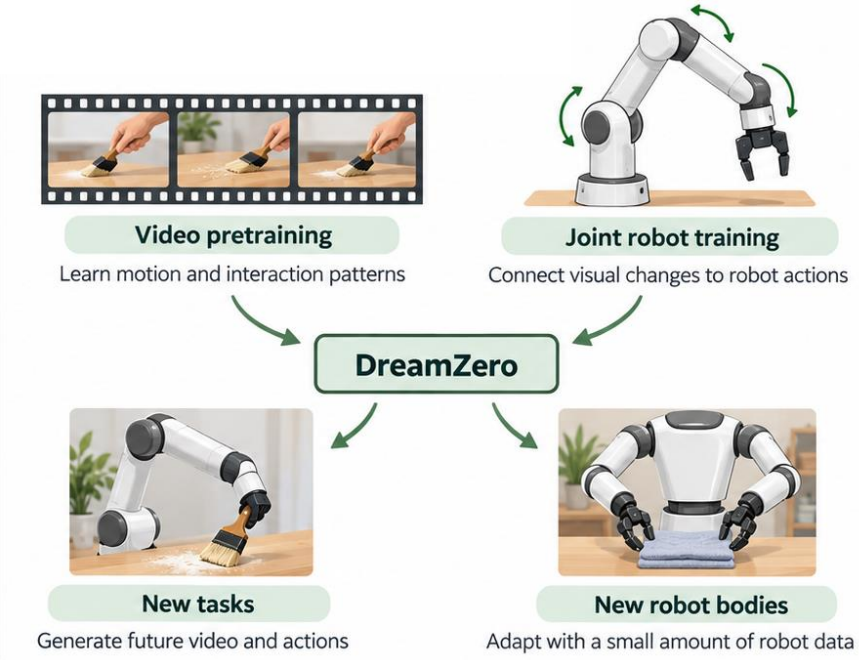
$$\mathcal{L} = \underbrace{\|\hat{v}_z - v_z\|^2}_{\text{Video velocity prediction error}} + \underbrace{\|\hat{v}_a - v_a\|^2}_{\text{Action velocity prediction error}}$$

  - Only video and action data, less reliance on repeating each task.
  - Learn from diverse robot demonstrations.

- **Transfer knowledge to new tasks and robot bodies**

DreamZero connects two sources of experience:
Video pretraining: Learn how objects move and interactions unfold.
Joint robot training: Learn how visual changes correspond to robot actions.

❖ Like an LLM



Pretraining

- **Training phase**



Wan2.1-I2V-14B-480

DreamZero (This paper)

- **Training data**

Robot data: Diversity over repetition

AgiBot G1: ~500 hours across 22 environments

Homes, restaurants, supermarkets, cafés, and offices

Each episode averages **4.4 minutes** and **~42 subtasks**

Franka: Separate training using the public DROID dataset

- **The aim:** Learn transferable skills from varied experience.

without running a physical robot during Pre-training.

Post-training: Practice Specific Tasks

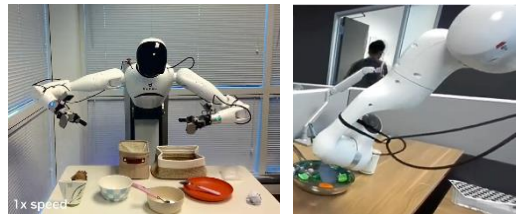
- **Training task**



TaskTraining dataGoal	Training time	Task
Shirt folding	33 hours	Fold a T-shirt through 5 stages
Fruit packing	12 hours	Pack 10 fruits into a bag
Table bussing	40 hours	Sort 5 trash items and 5 dishware items into separate bins

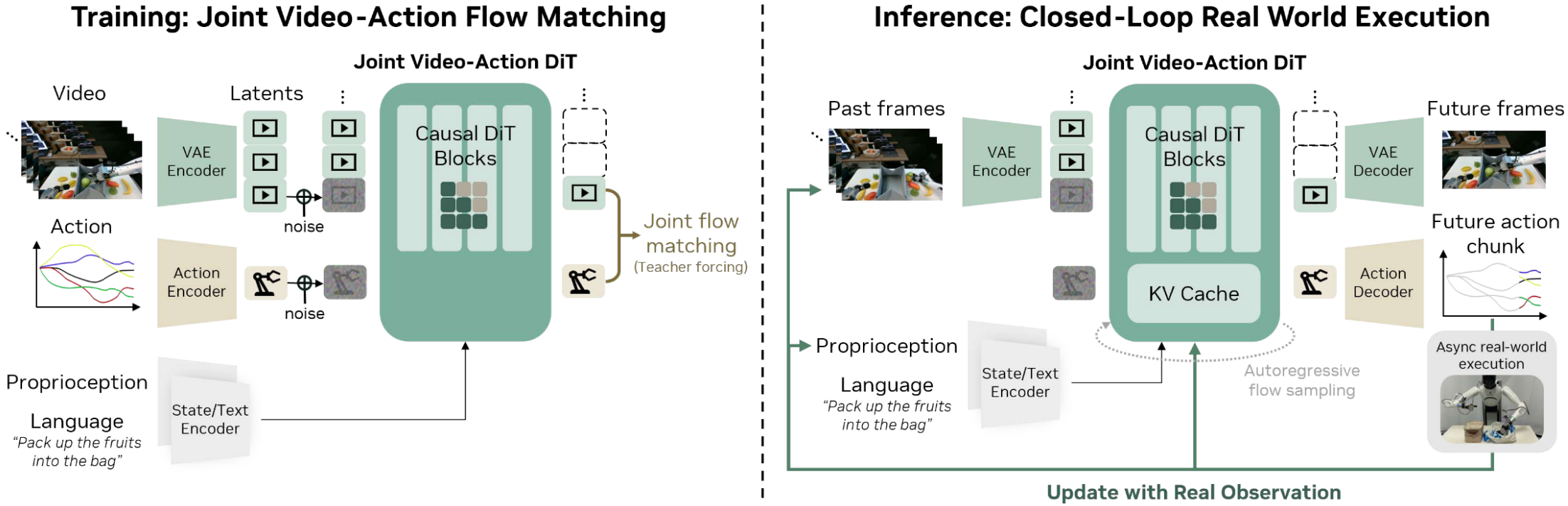
- During post-training, demonstrations are collected using a **physical robot**

  About 30 hours training data, a model can adapt to a new robot.

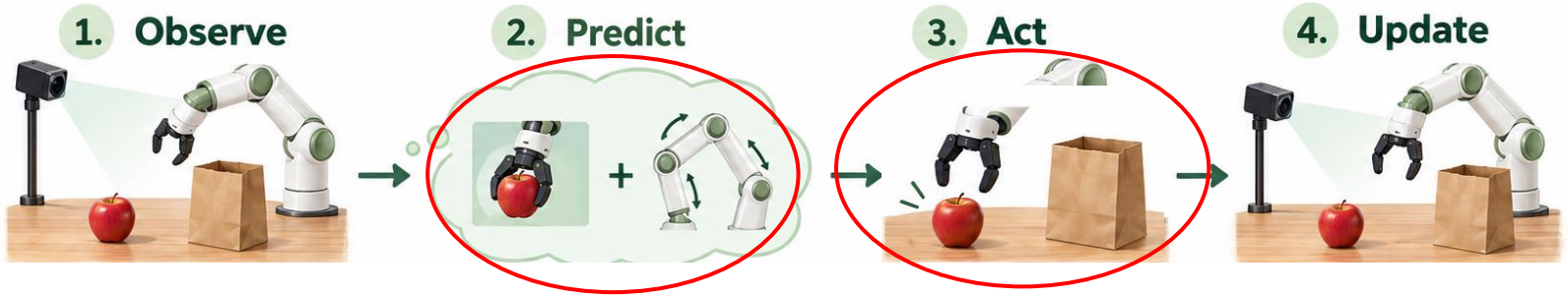


Inference: Predict, Act, and Update

➢ Inference Framework



➢ Why use real observation



The next prediction starts from what actually happened.

Results

Results

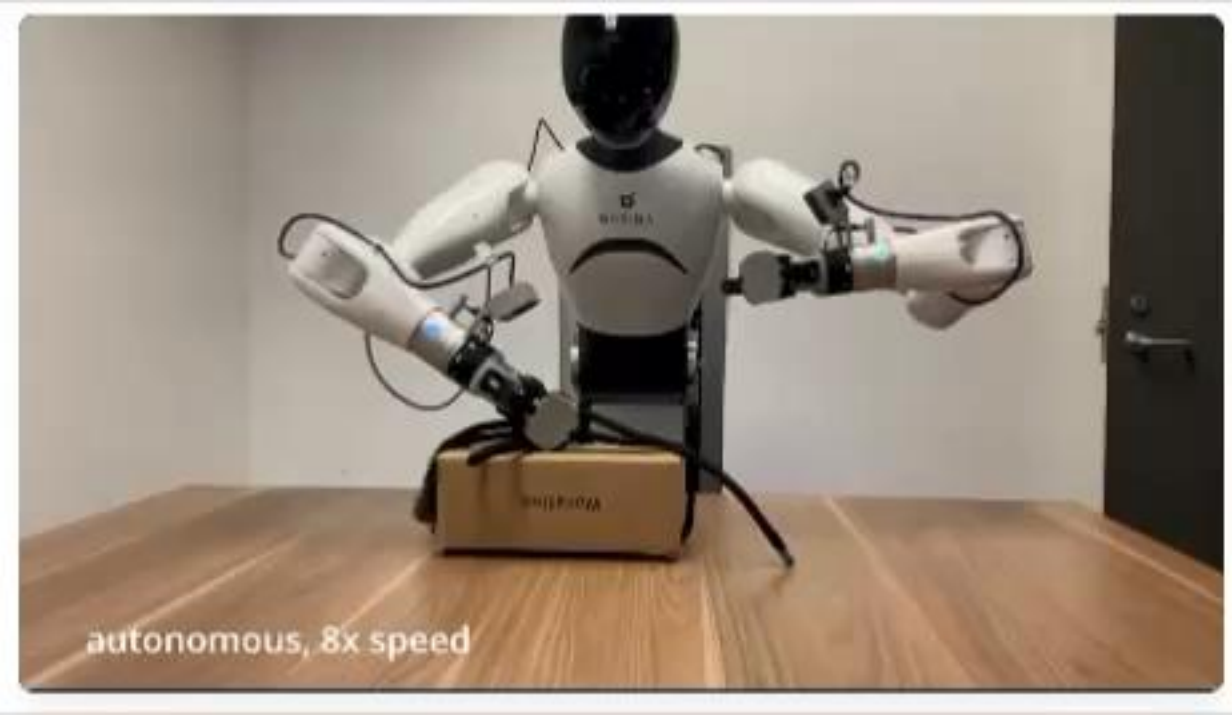
- **Post-training task**



Pick up the green pear on the table and place it on the blue checkered bowl.

Results

- **Unseen task**



Untie the knot of the package.

Results

- **Seen task & Unseen Verbs**



Remove the pair of gloves from the open drawer and put it on the table

Conclusion

➢ **Can a robot do a task it has never practiced?**

- **Yes, in some cases. DreamZero shows that this is possible.**

  - It combines knowledge from videos with robot demonstrations.
  - It jointly predicts future scenes and robot actions.
  - It can perform some tasks without task-specific robot demonstrations.

❖ **It has not practiced every task, but it can build on what it has learned.**

Thank you!

Happy to take questions.